Chloaelis

A dual-channel filter, tube and delay instrument with a Lorenz attractor, a flock of boids and two field recordings built in.

Manual for version 1.0.0



About this manual

This manual covers Chloaelis 1.0.0, the plug-in (AU, VST3, LV2) and the desktop app. It follows the README, with the full key to the symbols, a reference for every parameter your DAW can automate, and answers to common problems.



Spatial audio version: Chloaeilius. Chloaelis as a third-order Ambisonics encoder (16 channels, ACN/SN3D, for the IEM and SPARTA decoders): each side's dry sound is a point, its main echo rides the Lorenz attractor and its poly taps ride a 3-D flock, so the echoes fly round you, with distance and Doppler and a sphere you can turn to see where everything is. It lives in the `Chloaeilius` folder of this repository.



What it is

Chloaelis is two complete signal paths, hard-panned: the **left input goes through the left side to the left output**, and the right through the right. Each side runs

input > multimode filter > tube > delay > output

(or delay > tube > filter, with the order switch). Shared controls in the middle tie the two sides together:

- **Filter**: continuous low pass > band pass > high pass > notch, with resonance.
- **Tube**: four types, each oversampled 4x: I 12AX7 (the default: warm, two triode stages), II EL84 (push-pull pentodes, crunchy), III 6L6 (with power-supply sag, it blooms) and IV starved (a low-voltage 12AX7, gated and sputtery).
- **Delay engines**: *tape* (an ageing tape loop: hiss, wow and flutter, oxide wear; it ages while it runs and *refresh* puts on new tape), *PT2399* (the delay chip, gritty at long times) and *BBD* (bucket brigade, dark and crunchy). Up to 1 s, regen that self-oscillates, a feedback high-pass, freeze, random freeze, and slew (pitch-bending) or fade time changes.
- **Poly**: fades in up to 8 delay taps per side at irregular offsets. All the way up it becomes a reverb-like cloud. **Flock** lets a boids simulation move those taps' times, so they drift, cluster and scatter.
- **Lorenz attractor**: its x and y move the left and right cutoffs.
- **LFOs**: triangle LFOs on filter mode, delay time and tube drive, per side.
- **Master feedback**, crossed over: the left output feeds the right input and the right output the left.
- **Two built-in recordings** to play through it: a long loop from *baserecording1* and a hydrophone recording. Fade them in and out; the left channel goes to the left side and the right to the right.
- Every switch (order, delay engine, time mode, tube type) **crossfades**: echoes ring out in the old setting while new sound goes through the new one, so there are no cuts.
- It **remembers** how you left it, and ↺ resets everything.

Download and install

Ready-made builds for all three systems are on the **Releases page**:

System	File	Contains
macOS	`Chloaelis-macOS.zip`	the app, the Audio Unit and the VST3 (Intel + Apple Silicon)
Windows	`Chloaelis-Windows.zip`	the app (`Chloaelis.exe`) and the VST3
Linux	`Chloaelis-Linux.tar.gz`	the app, the VST3 and the LV2

`SHA256SUMS.txt` on the same page lists checksums if you want to verify a download.

> The builds aren't signed by Apple or Microsoft (that costs money), so the first launch needs one extra step on macOS and Windows. It's described below.

macOS (Homebrew)

If you use Homebrew:

```
brew install --cask femifleming/chloaelis/chloaelis
```

This installs the app into `/Applications` and the Audio Unit and VST3 into `~/Library/Audio/Plug-Ins`. To update later: `brew upgrade --cask chloaelis`. To remove: `brew uninstall --cask chloaelis`.

macOS (manual)

1. Download `Chloaelis-macOS.zip` from the Releases page and unzip it.
2. Drag **Chloaelis.app** into **Applications**.
3. For the plug-ins, open Finder, press ⇧⌘**G** and go to `~/Library/Audio/Plug-Ins`. Copy **Chloaelis.component** into the `Components` folder and **Chloaelis.vst3** into `VST3` (create either folder if it's missing).
4. First launch: because the build isn't notarised, macOS will refuse to open it the first time. Either right-click the app and choose **Open**, then **Open** again, or run this once in Terminal: ```sh xattr -dr com.apple.quarantine /Applications/Chloaelis.app ~/Library/Audio/Plug-Ins/Components/Chloaelis.component ~/Library/Audio/Plug-Ins/VST3/Chloaelis.vst3 ```
5. Restart your DAW (Logic, Ableton Live, Reaper and so on) so it rescans its plug-ins. Chloaelis appears under the maker **Chloaelis**.

Windows

1. Download `Chloaelis-Windows.zip` and unzip it.
2. **App**: put `Chloaelis.exe` anywhere and double-click it. If SmartScreen says "Windows protected your PC", click **More info** > **Run anyway** (it only asks once).
3. **Plug-in**: copy the `Chloaelis.vst3` folder into `C:\Program Files\Common Files\VST3`.
4. Restart your DAW so it rescans.

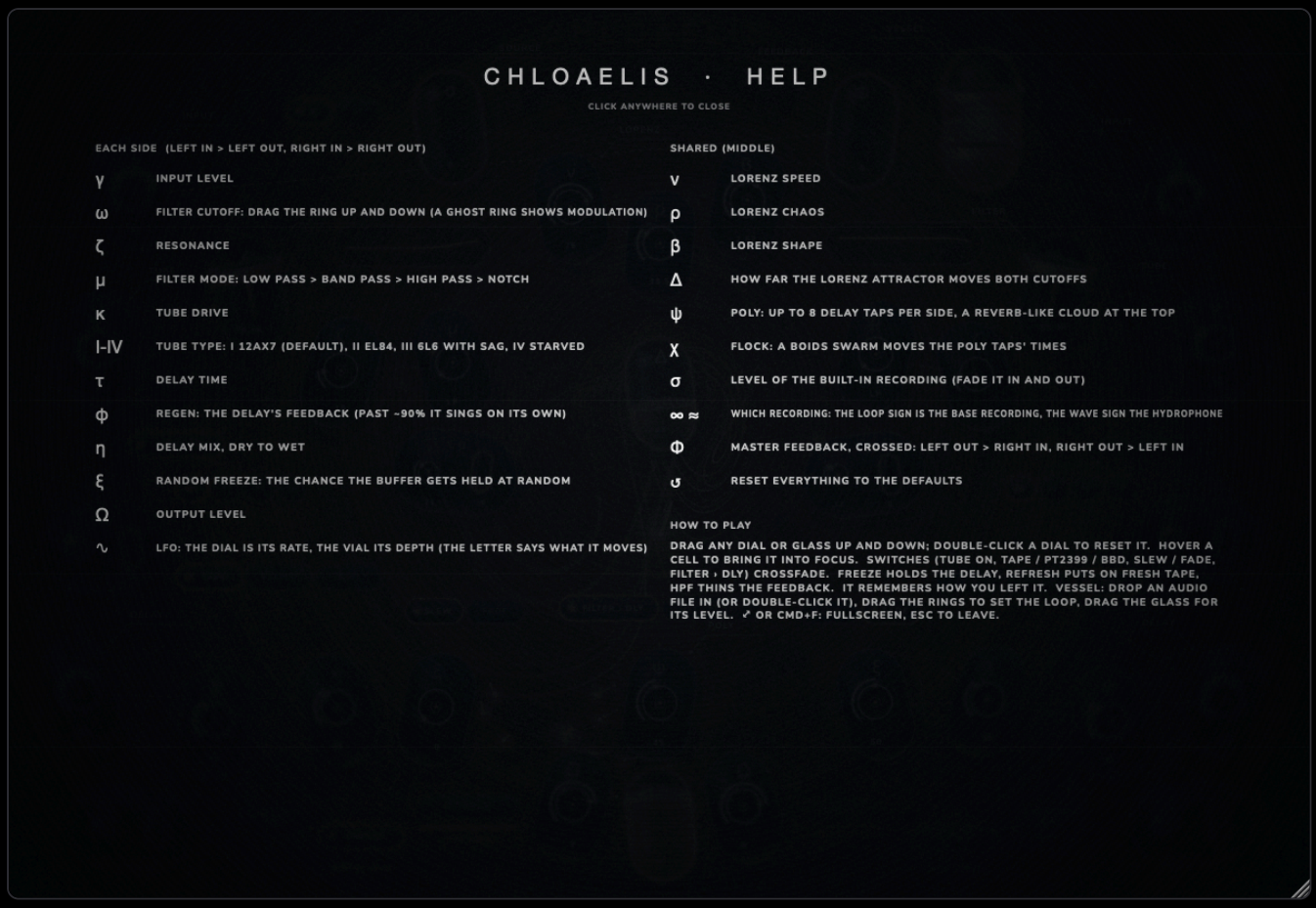
Linux

1. Download `Chloaelis-Linux.tar.gz` and unpack it: ```sh tar -xzf Chloaelis-Linux.tar.gz ```
2. **App**: run `./Chloaelis/Chloaelis` (you may want to copy it to `~/.local/bin`).
3. **Plug-ins**: ```sh mkdir -p ~/.vst3 ~/.lv2 cp -r Chloaelis/Chloaelis.vst3 ~/.vst3/ cp -r Chloaelis/Chloaelis.lv2 ~/.lv2/ ```

4. It needs ALSA or JACK, X11 and OpenGL, which a normal desktop already has.

How to use it

The layout



- The **left half** of the window is the left channel and the **right half** is the right channel. They mirror each other: input at the top, then filter, tube and LFOs, delay at the bottom, output at the bottom middle.
- The **middle** holds the shared controls: the Lorenz attractor (top), the poly and flock (bottom), the order switch, and the built-in recordings, master feedback and the vessel (top row).
- **Glowing dashed lines** show where signal and modulation flow. Pulses of light follow the signal.
- Cells **glow with level**: the side's colour when it's moving, warm white when it's hot and red when it's close to clipping.
- The **Lorenz attractor** floats in the background as a hologram, with dashed lines to the two cutoffs. The **boids** of the flock fly around the poly controls.

The symbols

Every control is named by a Greek letter. Press **?** (top right) for this list inside the app.

Each side

Symbol	Control	What it does
γ	Input	input level
ω	Cutoff	filter cutoff: drag the ring up and down. A faint ring shows where modulation has pushed it
ζ	Resonance	filter resonance

μ	Mode	filter mode: low pass > band pass > high pass > notch, continuously
κ	Drive	tube drive
I–IV	Tube type	I 12AX7 (default), II EL84, III 6L6 with sag, IV starved
TUBE ON		switches the tube in and out
τ	Time	delay time, 30 ms to 1 s
φ	Regen	the delay's feedback. Past about 90% it sings on its own
η	Mix	delay dry / wet
ξ	Random freeze	the chance that the delay buffer gets held for a while at random moments
TAPE / PT2399 / BBD		the delay engine
SLEW / FADE		time changes either glide (pitch bends) or crossfade (no pitch change)
HPF		high-pass filter in the delay feedback
FREEZE		holds the delay buffer. On tape, a held loop slowly wears away
REFRESH		puts on fresh tape (tape engine only)
Ω	Out	output level
∿	LFO	the dial is its rate, the small glass vial its depth. The letter next to ∿ says what it moves (μ mode, τ time, κ drive)

Shared (middle)

Symbol	Control	What it does
ν	Speed	how fast the Lorenz attractor moves
ρ	Chaos	the attractor's rho: low wanders gently, high is wild
β	Shape	the attractor's beta: reshapes the orbit
Δ	Depth	how far the attractor moves the two cutoffs (up to ±3.5 octaves)
FILTER › DLY / DLY › FILTER	Order	filter > tube > delay, or delay > tube > filter
ψ	Poly	up to 8 delay taps per side. All the way up is a reverb-like cloud
χ	Flock	how much the boids move the poly taps' times (up to about ±0.9 octave)
∞ / ≈	Source	which built-in recording: ∞ the base loop, ≈ the hydrophone
σ	Source level	fades the built-in recording in and out
Φ	Master feedback	crossed: left out > right in, right out > left in
↺	Reset	every control back to its default (the tape's wear is kept)

?	Help	the key to all of this, inside the app

Playing it

- **Drag up and down** on any dial, glass vial or ring to change it. **Double-click** a dial to reset it to its default.
- **Hover** over a control to bring it into focus. The depth of field follows you, and the field tilts slightly toward the cursor.
- **No input?** Turn up **σ** to play one of the built-in recordings through it, and pick which with **∞ / ≈**.
- **Ideas to start with**
  - *Ambient wash*: σ up, ψ (poly) and χ (flock) up, φ (regen) around 60%, tape engine.
  - *Lo-fi echo*: PT2399 or BBD, τ around 400 ms, φ around 70%, κ (drive) up a little.
  - *Moving filter*: Δ (Lorenz depth) up, ρ (chaos) up, ζ (resonance) up.
  - *Self-playing*: Φ (master feedback) towards 90% with some regen: the two sides feed each other and keep going on their own.
- Chloaelis **remembers its last state**: a new instance starts how the last one was closed. In a DAW, the project's own saved settings take priority.

The desktop app

The app (`Chloaelis.app`, `Chloaelis.exe` or `Chloaelis` on Linux) has a few extras:

- **Audio settings**: the **Options** button at the top left of the window chooses the audio device, the input and output channels and the buffer size.
- **Fullscreen**: the ⤢ button, or **⌘F** (Ctrl+F on Windows and Linux). It fills the whole screen with no menu bar or title bar, at any screen shape. **Esc** leaves it.
- **The vessel** (top right): a glass you can pour your own audio into.
  - **Drop an audio file** onto it (WAV, AIFF, FLAC, MP3, M4A, OGG or CAF, up to 10 minutes), or **double-click** it to choose one.
  - Its waveform runs up the inside, start at the bottom.
  - The **two rings** are the loop start and end: drag them.
  - **Drag the glass** itself (away from the rings) to set its level, shown by the liquid.
  - It plays into both sides like the built-in recordings, and is remembered with the rest of the state.

Help inside the app

Press **?** (top right) for a full key to the symbols and a short how-to. Click anywhere, or press Esc in the app, to close it.

Build from source

You need CMake 3.22 or newer and a C++20 compiler (Xcode on macOS, Visual Studio 2022 on Windows, GCC or Clang on Linux).

```
git clone --recursive https://github.com/femifleming/Chloaelis.git
cd Chloaelis
cmake -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build --config Release --target Chloaelis_All
```

The results are in `build/Chloaelis_artefacts/Release/` (`Standalone`, `VST3`, plus `AU` on macOS and `LV2` on Linux). Add `-DCHLOAELIS_COPY_AFTER_BUILD=ON` to have the plug-ins copied into your user plug-in folders after each build.

On Linux, install the development packages first (Debian / Ubuntu names):

```
sudo apt-get install libasound2-dev libjack-jackd2-dev libfreetype-dev libfontconfig1-dev \
  libx11-dev libxcomposite-dev libxcursor-dev libxext-dev libxinerama-dev libxrandr-dev \
  libxrender-dev libxi-dev libegl-dev libgl1-mesa-dev libglu1-mesa-dev mesa-common-dev ladspa-
sdk
```

If you cloned without `--recursive`, run `git submodule update --init` to fetch JUCE.

Tests (macOS): `cmake -B build -DCHLOAELIS_BUILD_TESTS=ON` builds `ChloaelisTest` (offline DSP checks) and `ChloaelisHostTest` (the full plug-in, headless; `CHL_AUDIOTEST=1` runs the audio checks).

Credits and licence

- Design and recordings: **Femi Fleming**. The two built-in recordings are Femi Fleming's field recordings (*baserecording1* and *hydrophone, June 2026, 3*).
- Inspired by the 2012 game *Splice* and the *Chloe* Eurorack module by Bizarre Jezabel.
- Built with JUCE 9.0.5, used under the GNU AGPLv3.
- Interface font: Nunito (a bold cut), SIL Open Font License 1.1 (see `Resources/fonts/OFL-Nunito.txt`).

Chloaelis is free software: you can redistribute it and/or modify it under the terms of the **GNU Affero General Public License v3** as published by the Free Software Foundation. See LICENSE.

Parameter reference

Every control your DAW can automate, by the name it shows there. The two sides have the same controls, so the right side's table follows the left's. Defaults are what a fresh instance starts with (reset brings them back).

Shared (the middle)

Name in your DAW	ID	Range	Default
Lorenz Speed	`lz_speed`	0% to 100%	35%
Lorenz Chaos	`lz_chaos`	ρ 24.5 to ρ 180.0	ρ 38.5
Lorenz Shape	`lz_shape`	β 0.50 to β 4.00	β 2.67
Lorenz > Cutoff	`lz_depth`	0% to 100%	0%
Poly Width	`width`	OFF to 100%	OFF
Poly Flock	`poly_flock`	0% to 100%	0%
Order	`order`	Filter > Delay, Delay > Filter	Filter > Delay
Source	`src_sel`	Base Loop, Hydrophone	Base Loop
Source Level	`src_level`	0% to 100%	0%
Master Feedback	`master_fb`	0% to 100%	0%

Left side

Name in your DAW	ID	Range	Default
L In Att	`l_inatt`	silent to 12.0 dB	0.0 dB
L Cutoff	`l_cutoff`	20 Hz to 20.0 kHz	20.0 kHz
L Resonance	`l_res`	0% to 100%	15%
L Mode	`l_morph`	LP to Notch	LP
L Drive	`l_drive`	0% to 100%	30%
L Time	`l_time`	30 ms to 1000 ms	330 ms
L Regen	`l_regen`	0% to 115%	40%
L Wet	`l_wet`	0% to 100%	35%
L Out	`l_out`	silent to 12.0 dB	0.0 dB
L Random Freeze	`l_randfrz`	0% to 100%	0%
L LFO>Mode Rate	`l_lfomode_rate`	0.02 Hz to 20.0 Hz	0.21 Hz
L LFO>Mode Depth	`l_lfomode_depth`	0% to 100%	0%
L LFO>Time Rate	`l_lfotime_rate`	0.02 Hz to 20.0 Hz	0.11 Hz
L LFO>Time Depth	`l_lfotime_depth`	0% to 100%	0%
L LFO>Drive Rate	`l_lfodrive_rate`	0.02 Hz to 20.0 Hz	0.37 Hz
L LFO>Drive Depth	`l_lfodrive_depth`	0% to 100%	0%
L Delay HPF	`l_hpf`	off / on	Off
L Freeze	`l_freeze`	off / on	Off
L Tube On	`l_tubeon`	off / on	On

L Delay Type	`l_dtype`	Tape, PT2399, BBD	PT2399
L Time Change	`l_tmode`	Slew, Fade	Slew
L Tube Type	`l_tubetype`	12AX7, EL84, 6L6, Starved	12AX7

Right side

Name in your DAW	ID	Range	Default
R In Att	`r_inatt`	silent to 12.0 dB	0.0 dB
R Cutoff	`r_cutoff`	20 Hz to 20.0 kHz	20.0 kHz
R Resonance	`r_res`	0% to 100%	15%
R Mode	`r_morph`	LP to Notch	LP
R Drive	`r_drive`	0% to 100%	30%
R Time	`r_time`	30 ms to 1000 ms	410 ms
R Regen	`r_regen`	0% to 115%	40%
R Wet	`r_wet`	0% to 100%	35%
R Out	`r_out`	silent to 12.0 dB	0.0 dB
R Random Freeze	`r_randfrz`	0% to 100%	0%
R LFO>Mode Rate	`r_lfomode_rate`	0.02 Hz to 20.0 Hz	0.17 Hz
R LFO>Mode Depth	`r_lfomode_depth`	0% to 100%	0%
R LFO>Time Rate	`r_lfotime_rate`	0.02 Hz to 20.0 Hz	0.13 Hz
R LFO>Time Depth	`r_lfotime_depth`	0% to 100%	0%
R LFO>Drive Rate	`r_lfodrive_rate`	0.02 Hz to 20.0 Hz	0.29 Hz
R LFO>Drive Depth	`r_lfodrive_depth`	0% to 100%	0%
R Delay HPF	`r_hpf`	off / on	Off
R Freeze	`r_freeze`	off / on	Off
R Tube On	`r_tubeon`	off / on	On
R Delay Type	`r_dtype`	Tape, PT2399, BBD	PT2399
R Time Change	`r_tmode`	Slew, Fade	Slew
R Tube Type	`r_tubetype`	12AX7, EL84, 6L6, Starved	12AX7

Problems and answers

macOS says it can't be opened, or the DAW doesn't list it

The builds aren't notarised. Run `xattr -dr com.apple.quarantine` on the app and the plug-ins (see Installing), then rescan in your DAW.

There's no sound and nothing on the input

Turn up σ (source level) to play a built-in recording through it, and pick which with ∞ / ≈. In the desktop app, check Options for the audio device.

It keeps sounding after I stop

Regen past about 90 % sings on its own, and master feedback (Φ) keeps the two sides feeding each other. Turn them down, or reset with ↺.

A new instance doesn't start from the defaults

It remembers its last state, so a new instance starts how the last one was closed. A saved project still opens as you saved it. ↺ resets everything (the tape's wear is kept).